

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version C — SOLUTIONS

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Fall 2026 • Total: 100 points

About Version C. This additional practice exam and solution set were generated with GPT-5.6 Sol in Codex. They use the current Midterm 1 coverage and target about thirty minutes of work in a fifty-minute period. Please report any error, ambiguity, out-of-coverage question, or item that takes much longer than its point value suggests.

Problem 1. (30 points) **True or false; prove or disprove.**

- (a) **FALSE.** Since $\gcd(10, 28) = 2$ divides 6, the congruence has exactly two solutions modulo 28, not four. Dividing by 2 gives $5x \equiv 3 \pmod{14}$. The inverse of 5 modulo 14 is 3, so $x \equiv 9 \pmod{14}$; the two solutions modulo 28 are $x \equiv 9, 23$.
- (b) **TRUE.** The displayed cycles move exactly the seven points 1, 2, 3, 4, 5, 6, 7, so the remaining points 8 and 9 are fixed.
- (c) **FALSE.** The rule is not well defined: $\bar{0} = \bar{8}$ in $\mathbb{Z}/8\mathbb{Z}$, but the proposed images are $\bar{0}$ and $\bar{16} = \bar{4}$, which are different in $\mathbb{Z}/12\mathbb{Z}$.
- (d) **TRUE.** The identity matrix has determinant 1. If A and B have determinant 1, then $\det(AB^{-1}) = \det(A)\det(B)^{-1} = 1$, so the subgroup criterion applies.
- (e) **TRUE.** The identity lies in both H and K , so the intersection is nonempty. If $a, b \in H \cap K$, then ab^{-1} lies in both H and K , hence in $H \cap K$. The subgroup criterion applies.
- (f) **TRUE.** Let $G = \langle x \rangle$ and let $L \leq G$. The trivial subgroup is cyclic. Otherwise choose the least positive n with $x^n \in L$. If $x^m \in L$, write $m = qn + r$ with $0 \leq r < n$. Then $x^r = x^m(x^n)^{-q} \in L$, so minimality forces $r = 0$. Hence every element of L is a power of x^n , and $L = \langle x^n \rangle$.

Grading note. Each item is worth two points for the correct truth value and three for a valid reason. A bare circle earns only two points. In (a), the gcd criterion gives full reason-credit; listing the two solution classes is optional. In (c), recognizing that the rule is not well defined is the essential point. In (e), the reason must check the intersection with the subgroup criterion. In (f), the theorem may be quoted for full credit if it has already been proved in class.
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Problem 2. (25 points) **The unit group modulo 20.**

(a) The classes relatively prime to 20 are

$$U = \{\bar{1}, \bar{3}, \bar{7}, \bar{9}, \bar{11}, \bar{13}, \bar{17}, \bar{19}\}.$$

(b) We compute modulo 20:

u	useful powers	$ u $
$\bar{3}$	$\bar{3}^2 = \bar{9}, \quad \bar{3}^4 = \bar{1}$	4
$\bar{9}$	$\bar{9}^2 = \bar{1}$	2
$\bar{11}$	$\bar{11}^2 = \bar{1}$	2
$\bar{19}$	$\bar{19}^2 = \bar{1}$	2

(c) The group U has eight elements, so if it were cyclic it would have an element of order 8. But the square of every element of U is either $\bar{1}$ or $\bar{9}$, and $\bar{9}^2 = \bar{1}$. Thus every element satisfies $u^4 = \bar{1}$, so no element has order 8. Therefore U is not cyclic.

(d) First

$$\langle \bar{3} \rangle = \{\bar{1}, \bar{3}, \bar{9}, \bar{7}\}.$$

Multiplying these four elements by $\bar{11}$ gives the four additional elements

$$\bar{11}, \bar{13}, \bar{19}, \bar{17}.$$

Consequently

$$\langle \bar{3}, \bar{11} \rangle = \{\bar{1}, \bar{3}, \bar{7}, \bar{9}, \bar{11}, \bar{13}, \bar{17}, \bar{19}\} = U.$$

Grading note. In (a), deduct one point for each missing or nonunit class, up to four points. In (b), award two points per order when supported by a power calculation. In (c), “not cyclic” without ruling out order 8 earns only two points. In (d), four points are for the powers of $\bar{3}$, two for the four additional products obtained using $\bar{11}$, and two for concluding that all of U is generated.

Problem 3. (25 points) **Computing in D_{10} .**

(a) Using $sr^i s = r^{-i}$,

$$(sr^2)(sr^4) = sr^2 sr^4 = r^{-2} r^4 = r^2.$$

Also

$$(r^3 s)^{-1} = s^{-1} r^{-3} = sr^2.$$

(b) The identity has order 1. Since 5 is prime, each of r, r^2, r^3, r^4 has order 5. Finally,

$$(sr^i)^2 = sr^i sr^i = r^{-i} r^i = 1,$$

so each of s, sr, sr^2, sr^3, sr^4 has order 2.

(c) The subgroup contains

$$(sr)(sr^3) = sr sr^3 = r^{-1} r^3 = r^2.$$

Because $\gcd(2, 5) = 1$, $\langle r^2 \rangle = \langle r \rangle$, so the subgroup contains r . It also contains sr , and hence it contains $(sr)r^{-1} = s$. Therefore it contains the generators r and s of D_{10} , and $\langle sr, sr^3 \rangle = D_{10}$.

(d) The complete list is

$$\{1\}, \quad \langle r \rangle, \quad \langle s \rangle, \langle sr \rangle, \langle sr^2 \rangle, \langle sr^3 \rangle, \langle sr^4 \rangle, \quad D_{10}.$$

To see completeness, let $L \leq D_{10}$. If L contains a nonidentity rotation r^j , then $\langle r^j \rangle = \langle r \rangle$ because 5 is prime. Thus either $L = \langle r \rangle$, or, if L also contains a reflection, L contains both r and s and is all of D_{10} . If L contains no nonidentity rotation, it cannot contain two distinct reflections, since $(sr^i)(sr^j) = r^{j-i}$ is then a nonidentity rotation. Hence such an L is either trivial or one of the five order-2 reflection subgroups listed above.

Grading note. In (a), each computation is worth three points; require normal form. In (b), two points each go to the identity, the four nonidentity rotations, and the five reflections, with their orders justified. In (c), the key steps are producing r^2 , observing that it generates all rotations, and then producing s . In (d), four points are for the complete list and two for the argument that two distinct reflections force a nonidentity rotation. No use of Lagrange's Theorem is needed or intended.

Problem 4. (20 points) **Subgroups inside $\mathbb{Z}/24\mathbb{Z}$.**

- (a) The set H is nonempty because $\bar{0} \in H$. If $\bar{a}, \bar{b} \in H$, then $4\bar{a} = 4\bar{b} = \bar{0}$, and therefore

$$4(\bar{a} - \bar{b}) = 4\bar{a} - 4\bar{b} = \bar{0}.$$

Thus $\bar{a} - \bar{b} \in H$, so the subgroup criterion shows that $H \leq \mathbb{Z}/24\mathbb{Z}$.

- (b) The condition $4\bar{x} = \bar{0}$ means $24 \mid 4x$, or equivalently $6 \mid x$. Hence

$$H = \{\bar{0}, \bar{6}, \bar{12}, \bar{18}\} = \langle \bar{6} \rangle.$$

This cyclic group has order 4, so its generators are $\bar{6}$ and $\bar{18}$.

- (c) Both $\bar{8}$ and $\bar{18}$ are multiples of $\bar{2}$, so $K \subseteq \langle \bar{2} \rangle$. Conversely,

$$\bar{2} = \bar{18} - \bar{28} \in K,$$

so $\langle \bar{2} \rangle \subseteq K$. Therefore $K = \langle \bar{2} \rangle$, the subgroup of all even residue classes. Every element of H is even, so $H \subseteq K$ and

$$H \cap K = H = \{\bar{0}, \bar{6}, \bar{12}, \bar{18}\}.$$

Grading note. In (a), award two points for nonemptiness and five for a correct subtraction-criterion argument. In (b), three points are for the list and two for all generators. In (c), each containment is worth three points; the identity $\bar{2} = \bar{18} - \bar{28}$ is the intended Bézout-style step. The final intersection is worth two points.